

100 Days of Azure DevOps

Day 39 Handout — Promote the artifact, not the vibes. dependsOn is the chain. Prod does not get a second compile

4 - Continuous Delivery · Personal lab only · Educational content · Not a sales pitch

Architecture A — one run, one drop, three environments

Stage	Job type	Consumes
CI (or reuse resources.pipelines)	job: Build + PublishPipelineArtifact@1	Source. Produces artifact drop
Dev	deployment: environment: dev	download: current artifact: drop
Staging	deployment: dependsOn: Dev	The same drop. Different app settings
Prod	deployment: environment: prod dependsOn: Staging	The same drop. Approvals check from Day 38
Config	App settings / variable groups per env	Not a new zip. Bits stay identical

Architecture B — promotion rules

Rule	YAML / practice	Violation
Build once	Publish in CI only	dotnet publish / npm run build in Prod
Chain	Staging dependsOn: Dev; Prod dependsOn: Staging	dependsOn: [] on Prod because 'we are late'
Brake on pain	Checks on prod Environment	Approvals on Dev that teach people to click through
Budget	Prod may echo if no App Service today	Rebuilding 'just to have a real prod'

Architecture C — pitfalls that invent 'works in staging' ghosts

Pitfall	What you see	Fix
Rebuild per env	Staging Monday; Prod Tuesday; different transitive dep	download the drop. Never restore again
download: none	Deploy job compiles from checkout	download: current (same run) or download: ci
Config baked in zip	Staging connection string shipped to Prod	Slot settings / env vars; bits stay dumb
Skipped Staging	Prod waited on Dev only	dependsOn: Staging even when Staging is echo

One-liner to remember

Promote the artifact, not the vibes — rebuilds between envs invent 'works in staging' ghosts.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure DevOps org only. Project azure-100-labs. Environments dev and prod already exist. Add staging. Personal Azure optional.

- 1 Pipelines → Environments → New → staging (empty). Keep prod's Approvals check from Day 38.
- 2 Write pipelines/multi-env.yml: stages Dev → Staging → Prod with dependsOn chain. Each is a deployment job.
- 3 CI publishes drop. Each env stage downloads it (download: current artifact: drop). No compile/publish in those stages.
- 4 Dev and Staging can echo if you are not spending. Prod must wait on the environment: prod approval.
- 5 Run once. Graph should show Dev, then Staging, then waiting on Approve — not three independent builds.
- 6 If any stage runs npm/dotnet/mvn, delete those steps. Config differences belong in variable groups, not a second artifact.

Starter YAML — three stages, one drop, approval on prod

```
# pipelines/multi-env.yml
trigger: none
stages:
- stage: CI
  jobs:
  - job: Build
    pool: { vmImage: ubuntu-latest }
    steps:
    - script: echo "CI bits" > $(Build.ArtifactStagingDirectory)/app.txt
    - task: PublishPipelineArtifact@1
      inputs: { targetPath: $(Build.ArtifactStagingDirectory), artifact: drop }
- stage: Dev
  dependsOn: CI
  jobs:
  - deployment: DeployDev
    environment: dev
    strategy:
      runOnce:
        deploy:
          steps:
          - download: current
            artifact: drop
          - script: echo "Dev gets $(Pipeline.Workspace)/drop – no rebuild"
- stage: Staging
  dependsOn: Dev
  jobs:
  - deployment: DeployStaging
    environment: staging
    strategy:
      runOnce:
        deploy:
          steps:
          - download: current
            artifact: drop
          - script: echo "Staging gets the same drop"
- stage: Prod
  dependsOn: Staging
  jobs:
  - deployment: DeployProd
    environment: prod
    strategy:
      runOnce:
        deploy:
          steps:
          - download: current
            artifact: drop
          - script: echo "Prod waits on Environment checks, then same drop"
```

Done checklist

- dependsOn chain is Dev → Staging → Prod
- Every env stage downloads drop; none rebuild
- Prod waited on the Approvals check
- Can explain why a second compile is a second dice roll

Tomorrow — Day 40

Phase 4 mini project — green path with visible brakes, same artifact, recap without heroics.

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